

ECx00M-CN&ECx00N-CN

MUX Application Note

LTE Standard Module Series

Version: 1.1

Date: 2023-07-12

Status: Released



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About the Document

Revision History

Version	Date	Author	Description
-	2021-04-12	Evin ZHU	Creation of the document
1.0	2022-10-31	Evin ZHU	First official release
1.1	2023-07-12	Sawyer HE/ Evin ZHU	1. Added the applicable modules EC600M-CN and EC800M-CN. 2. Updated the note that different models support MUX feature differently due to the flash size of the module (Chapter 1.1). 3. Updated the supported range of <port_speed> in AT+CMUX (Chapter 3.3).

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1 Introduction

The document defines Quectel MUX (multiplexer) protocol between UE and TE and provides examples of how to use the MUX function on Quectel LTE Standard ECx00M-CN and ECx00N-CN family modules.

1.1. Applicable Modules

Table 1: Applicable Modules

Module Family	Model
ECx00M-CN	EC200M-CN
	EC600M-CN
	EC800M-CN
ECx00N-CN	EC200N-CN
	EC800N-CN

NOTE

Limited by the flash size of the module, different models of the modules support MUX feature differently. For more details, please contact Quectel Technical Support.

2 Overview of Quectel Multiplexing System

The multiplexer provides mechanisms for conveying data streams between TE and UE over a single physical port. Quectel multiplexing system creates four virtual channels on one physical port to transmit multiple streams of data simultaneously. It enables the physical port to act like four real physical channels for the application. Each virtual channel supports SMS, PPP dialing, etc.

All data from the application are packed into different frames which consist of the data and protocol fields that clearly indicate channel number, length of the information, FCS, and so on. The frames are transmitted as data streams via the serial port. After arriving at the other peer, the data are unpacked as four streams by the multiplexer protocol stack and transmitted to the application. In addition to the information field, the control signals are also simulated.

Each channel between TE and UE is called a DLC (Data Link Connection) and is established separately and sequentially.

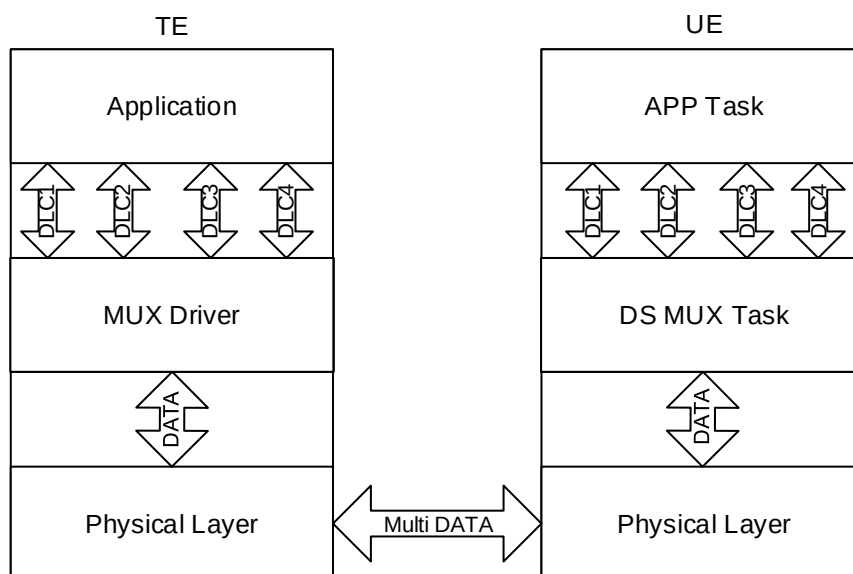


Figure 1: MUX Architecture

3 MUX AT Command

3.1. AT Command Introduction

3.1.1. Definitions

- **<CR>** Carriage return character.
- **<LF>** Line feed character.
- **<...>** Parameter name. Angle brackets do not appear on the command line.
- **[...]** Optional parameter of a command or an optional part of TA information response. Square brackets do not appear on the command line. When an optional parameter is not given in a command, the new value equals its previous value or the default settings, unless otherwise specified.
- **Underline** Default setting of a parameter.

3.1.2. AT Command Syntax

All command lines must start with **AT** or **at** and end with **<CR>**. Information responses and result codes always start and end with a carriage return character and a line feed character: **<CR><LF><response><CR><LF>**. In tables presenting commands and responses throughout this document, only the commands and responses are presented, and **<CR>** and **<LF>** are deliberately omitted.

Table 2: Types of AT Commands

Command Type	Syntax	Description
Test Command	AT+<cmd>=?	Test the existence of the corresponding command and return information about the type, value, or range of its parameter.
Read Command	AT+<cmd>?	Check the current parameter value of the corresponding command.
Write Command	AT+<cmd>=<p1>[,<p2>[,<p3>[...]]]	Set user-definable parameter value.
Execution Command	AT+<cmd>	Return a specific information parameter or perform a specific action.

3.2. Declaration of AT Command Examples

The AT command examples in this document are provided to help you familiarize with AT commands and learn how to use them. The examples, however, should not be taken as Quectel's recommendation or suggestions about how you should design a program flow or what status you should set the module into. Sometimes multiple examples may be provided for one AT command. However, this does not mean that there exists a correlation among these examples and that they should be executed in a given sequence.

3.3. AT+CMUX Multiplexing Mode

This command enables the multiplexing protocol control channel and sets parameters for the control channel.

AT+CMUX Multiplexing Mode	
Test Command AT+CMUX=?	Response +CMUX: (list of supported <transparency>s),(range of supported <subset>s),(range of supported <port_speed>s),(range of supported <N1>s),(range of supported <T1>s),(range of supported <N2>s),(range of supported <T2>s),(range of supported <T3>s),(range of supported <k>s) OK
Read Command AT+CMUX?	Response +CMUX: <transparency>[,<subset>[,<port_speed>[,<N1>[,<T1>[,<N2>[,<T2>[,<T3>[,<k>]]]]]]] OK
Write Command AT+CMUX=<transparency>[,<subset>[,<port_speed>[,<N1>[,<T1>[,<N2>[,<T2>[,<T3>[,<k>]]]]]]]	Response OK Or ERROR If there is an error related to ME functionality: +CME ERROR: <err>
Maximum Response Time	300 ms
Characteristics	The command takes effect immediately. The configuration is not saved.

Parameter

<transparency>	Integer type. Multiplexer transparency mechanism. 0 Basic option
<subset>	Integer type. Defines the way in which the multiplexer control channel is set up. A virtual channel may subsequently be set up differently but in the absence of any negotiation for the settings of a virtual channel, it shall be set up according to the control channel <subset> setting. 0 UIH frame used only 1 UI frame used only (Reserved) 2 I frame used only (Reserved)
<port_speed>	Integer type. Transmission rate of the physical port. 1 9600 bps 2 19200 bps 3 38400 bps 4 57600 bps 5 115200 bps 6 230400 bps 7 460800 bps 8 921600 bps
<N1>	Integer type. Maximum frame size. Range: 1–2048. Default value: 127. Unit: byte.
<T1>	Integer type. Acknowledgement timer, i.e., the time UE waits for an acknowledgement before resorting to another action (e.g., transmitting a frame). Range: 1–255. Default value: 10. Unit: ten milliseconds.
<N2>	Integer type. Maximum number of re-transmissions. Range: 0–100. Default value: 3.
<T2>	Integer type. Response timer for the multiplexer control channels. Range: 2–255. Default value: 30. Unit: ten milliseconds. <T2> must be longer than <T1>.
<T3>	Integer type. Wake up response timer. Range: 1–255. Default value: 10. Unit: second. (Not supported currently)
<k>	Integer type. Window size. Defines the maximum number of I frames that can be parsed by a DLC. (Not supported currently)
<err>	Error codes. See document [1] for details.

Example

```

AT+CMUX=0
OK
AT+CMUX?
+CMUX: 0,0,5,127,10,3,30,10,2
OK

```

NOTE

AT+CMUX can only be executed on UART1 since the modules only support multiplexer function on UART1.

4 Multiplexer Protocol

This chapter explains the technical details of the multiplexer protocol.

4.1. Frame Structure

All information transmitted between the TE and UE is conveyed in frames.

Table 3: MUX Frame Structure

Opening Flag	Address	Control	Length	Information	FCS	Closing Flag
1 octet	1 octet	1 octet	1–2 octet	Uncertain length	1 octet	1 octet

4.1.1. Flag Sequence Field

Each frame begins and ends with a flag sequence octet (0xF9).

4.1.2. Address Field

The address field consists of a single octet. It contains the DLCI (Data Link Connection Identifier), the C/R bit and the address field extension bit (EA) as shown in the table below.

Table 4: Address Field

Bit 1	Bit 2	Bit 3	Bit 4	Bit 5	Bit 6	Bit 7	Bit 8
EA	C/R						DLCI

4.1.2.1. DLCI

The DLCI identifies the virtual channel between TE and UE. Multiple DLCIs shall be supported but the number is implementation-specific. The DLCIs are dynamically assigned.

4.1.2.2. C/R

The C/R (command/response) bit identifies the frame as either a command or a response. TE sends a command to the UE with the C/R bit set to 1, and the UE responds with the C/R bit set to 1. The UE sends a command with the C/R bit set to 0, and the TE responds with the C/R bit set to 0.

Table 5: C/R Bit Usage

Command/Response	Direction	C/R Value
Command	TE → UE	1
	UE → TE	0
Response	UE → TE	1
	TE → UE	0

4.1.2.3. EA

EA bit extends the range of the address field. When the EA bit is set to 1 in an octet, it signifies that this octet is the last octet of the address field. When the EA bit is set to 0, it signifies that it is followed by another octet of the address field.

NOTE

EA can only be set to 1 currently.

4.1.3. Control Field

The content of the control field defines the frame type. See **Chapter 4.2** for more information about frame types.

Table 6: Coding of Control Field

Frame Type	HEX (P/F=0)	1	2	3	4	5	6	7	8
SABM (Set Asynchronous Balanced Mode)	0x2F	1	1	1	1	P/F	1	0	0
UA (Unnumbered Acknowledgement)	0x63	1	1	0	0	P/F	1	1	0
DM (Disconnected Mode)	0x0F	1	1	1	1	P/F	0	0	0
DISC (Disconnect)	0x43	1	1	0	0	P/F	0	1	0
UIH (Unnumbered Information with Header Check)	0xEF	1	1	1	1	P/F	1	1	1
UI (Unnumbered Information)	0x03	1	1	0	0	P/F	0	0	0

P/F is the Poll/Final bit. The poll (P) bit set to 1 shall be used by one station to solicit (poll) a response or sequence of responses from the other station.

The final (F) bit set to 1 shall be used by a station to indicate the response frame transmitted as the result of a soliciting (poll) command.

The poll/final (P/F) bit shall serve a function in both command frames and response frames. (In command frames, the P/F bit indicates a poll; in response frames, it indicates the final frame.)

- When DCE sends a message frame, P/F is set to 0.
- When DCE receives a message frame whose P/F is set to 1 from DLCI 0, DCE will give priority to responding to it and set response frame's P/F to 1.
- When DCE sends a control frame, P/F is set to 1.
- When DCE sends UIH frame via a DLC other than DLCI 0, P/F is set to 0.
- DCE only processes SABM and DISC frames whose P/F is set to 1.

4.1.4. Length Field

Table 7: The Structure of Length Field

Bit 1	Bit 2	Bit 3	Bit 4	Bit 5	Bit 6	Bit 7	Bit 8
EA	L1	L2	L3	L4	L5	L6	L7

L1 to L7 bits indicate the length of the following data field that contains fewer than 128 bytes.

Bit 1	Bit 2	Bit 3	Bit 4	Bit 5	Bit 6	Bit 7	Bit 8
0	L1	L2	L3	L4	L5	L6	L7

Bit 1	Bit 2	Bit 3	Bit 4	Bit 5	Bit 6	Bit 7	Bit 8
L8	L9	L10	L11	L12	L13	L14	L15

The length field may be extended by using the EA bit. When the EA bit is set to 1 in an octet, it signifies that this octet is the last octet of the length field. When the EA bit is set to 0, it signifies that it is followed by another octet of the length field. The total length of the length field is 15 bits in that case.

4.1.5. Information Field

The information field is the payload of the frame and carries the user data and any convergence layer information. The field is octet structured and only presents in UIH frames.

4.1.6. FCS Field

In the case of the UIH frame, the contents of the information field are not included in FCS calculation. FCS is calculated only based on the contents of the address, control, and length fields. This means that only the delivery to the correct DLCI is protected, but not the information.

4.2. Frame Types

4.2.1. SABM

SABM is a command frame and is used to establish DLC between TE and UE.

4.2.2. UA

The UA frame is a response to SABM or DISC frame, as illustrated in the following figure.

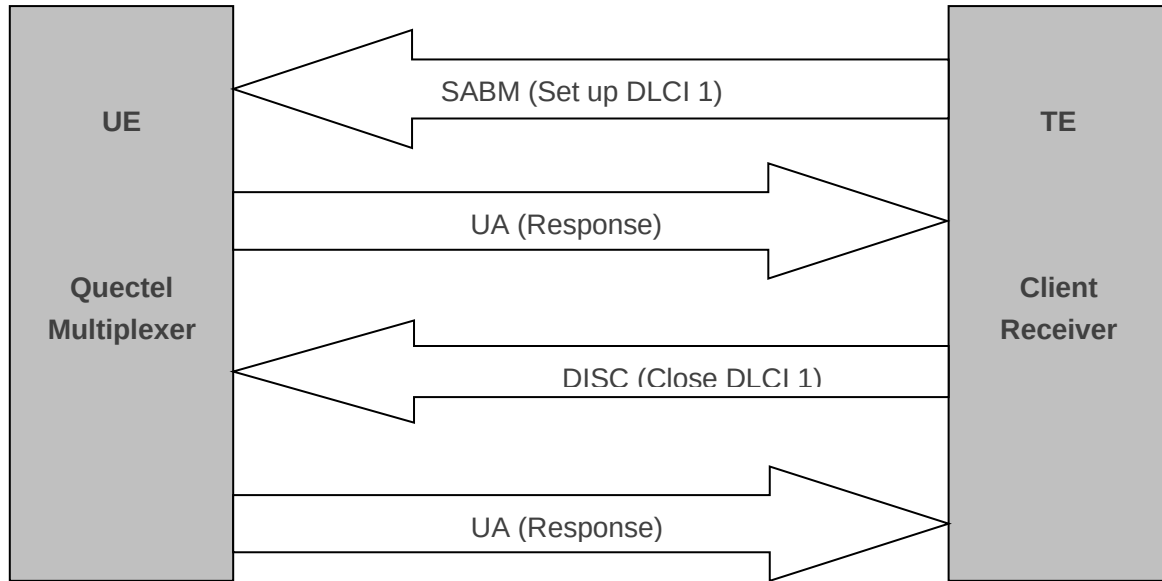


Figure 2: UA Frame (Response)

4.2.3. DM

The DM response frame is used to report a status when the station is logically disconnected from the data link. When in disconnected mode, no commands are accepted until the disconnected mode is terminated after receiving a SABM command. If a DISC command is received while in disconnected mode, a DM response is sent.

4.2.4. DISC

DISC is a command frame and is used to close down a DLC. Prior to executing the command, the receiving station confirms the acceptance of the DISC command with a UA response. See **Figure 2** for details.

4.2.5. UIH

The UIH command/response sends user data at either station.

4.3. Multiplexer Control Channel

At the initiation of communication between the TE and UE, a control channel is set up with DLCI 0. Control channel is the first channel established at the initiation of the multiplexer between the TE and UE and it has the DLCI value 0. This channel is used to convey information between the two multiplexers.

Multiplexer control channel is the basic channel used to establish a DLC, launch power saving, wake up from power saving and implement the flow control mechanism.

4.3.1. Message Format

All UIH frames are transmitted through the control channel and conform to the following type, length, and value format.

Table 8: Message Format

Type	Length	Value 1	Value 2	...	Value n
------	--------	---------	---------	-----	---------

Each box in the table above represents a field of minimum size per octet.

4.3.1.1. Type Field

Format of first type field octet:

Table 9: Message Format – Type Field

Bit 1	Bit 2	Bit 3	Bit 4	Bit 5	Bit 6	Bit 7	Bit 8
EA	C/R	T1	T2	T3	T4	T5	T6

The EA bit is an extension bit and is set to 1 in the last octet of the sequence. In other octets, EA is set to 0. In Quectel multiplexing system, only one octet is transmitted, therefore EA is always set to 1.

The C/R bit indicates whether the message is a command or a response.

The T bits indicate the type coding. Each command has a unique pattern of bit sequence. This means that a single-octet type field can encode 63 different message types. Only single octet message types are defined in this document.

4.3.1.2. Length Field

The structure of the length field octet:

Table 10: Message Format – Length Field

Bit 1	Bit 2	Bit 3	Bit 4	Bit 5	Bit 6	Bit 7	Bit 8
EA	L1	L2	L3	L4	L6	L6	L7

The EA bit is an extension bit. It is set to 1 in the last octet of the sequence. In other octets, EA is set to 0. In Quectel multiplexing system, only one octet is transmitted, so EA is always set to 1.

The L bits define the number of value octets that follow. L1 is the LSB and L7 is the MSB, which permits constructing messages with up to 127 value octets.

4.3.1.3. Value Field

The contents of the value octets are defined for each message type in **Chapter 4.3.2**.

4.3.2. Message Type and Actions

4.3.2.1. Multiplexer Close-Down (CLD)

The multiplexer close-down command is used to reset the link to normal AT command mode without multiplexing. The multiplexer close-down messages use the following type field octet:

Table 11: Multiplexer Close-Down Message – Type Field Format

Bit 1	Bit 2	Bit 3	Bit 4	Bit 5	Bit 6	Bit 7	Bit 8
EA	C/R	0	0	0	0	1	1

The length byte contains the value 0 and there are no value octets.

4.3.2.2. Flow Control on Command (FCon)

The flow control command is used to handle the aggregate flow. When either entity is able to receive new information, it transmits FCon command.

The length byte contains the value 0 and there are no value octets.

The format of the type field octet:

Table 12: Flow Control on Command – Type Field Format

Bit 1	Bit 2	Bit 3	Bit 4	Bit 5	Bit 6	Bit 7	Bit 8
EA	C/R	0	0	0	1	0	1

4.3.2.3. Flow Control off Command (FCoff)

The flow control command is used to handle the aggregate flow. When either entity is not able to receive information, it transmits the FCoff command. The opposite entity is not allowed to transmit frames except on the control channel (DLCI = 0).

The length byte contains the value 0 and there are no value octets.

The format of the type field octet:

Table 13: Flow Control off Command – Type Field Format

Bit 1	Bit 2	Bit 3	Bit 4	Bit 5	Bit 6	Bit 7	Bit 8
EA	C/R	0	0	0	1	1	0

4.3.2.4. Modem Status Command (MSC)

It is desired to convey virtual V.24 control signals to a data stream by sending the MSC command. The MSC has one mandatory control signal byte and an optional break signal byte. This command is only applicable when the basic option is chosen.

Send this command prior to any user data after a DLC is created.

Table 14: Modem Status Command

Command	Length	DLCI	V.24 Signals	Break Signals
---------	--------	------	--------------	---------------

The length byte contains the value 2 or 3 and there are 2 or 3 value octets.

Both the DTE and DCE use this command to notify each other of the status of their own V.24 control signals. MSC length is either 4 or 5 bytes depending on the break signal.

NOTE

Break signals are currently not supported.

The format of the command field octet:

Table 15: Modem Status Command – Command Field Format

Bit 1	Bit 2	Bit 3	Bit 4	Bit 5	Bit 6	Bit 7	Bit 8
EA	C/R	0	0	0	1	1	1

The C/R bit indicates if it is a Modem Status Command or a Modem Status Response.

Every time the signals change, the DTE or DCE sends this command to indicate the current status of each signal. When a DTE or DCE receives a Modem Status Command, it always sends a Response back. The mappings of the V.24 signals to the bits in the control signal octet for the receiver and sender are given in **Table 18** and **Table 19** below, respectively.

In an MSC command, the status of the sender's own V.24 signals is sent. However, in a Response, a copy of the V.24 signals is received from the Command frame that is returned.

The DLCI field identifies the specific DLC to which the command applies. Bit 2 is always set to 1 and the EA bit is set according to the description in **Chapter 4.1.2.3**.

Table 16: Format of Address Field

Bit 1	Bit 2	Bit 3	Bit 4	Bit 5	Bit 6	Bit 7	Bit 8
EA	1				DLCI		

The DLCI field is followed by the control signal field that contains a representation of the signal state in the format shown below. The use of the extension bit allows adding other octets to cater for other circumstances. At present, an optional second octet is defined for handling the transmission of break signals.

Table 17: Format of Control Signal Octet

Bit No.	Bit 1	Bit 2	Bit 3	Bit 4	Bit 5	Bit 6	Bit 7	Bit 8
Signal	EA	FC	RTC	RTR	Reserved (0)	Reserved (0)	IC	DV

Description of the control signal byte:

Bit 1: The EA bit is set to 1 in the last octet of the sequence; in other octets, EA is set to 0. In Quectel multiplexing system, only one octet is transmitted, so EA is always set to 1.

Bit 2: Flow Control (FC). The bit is set to 1 when the device is unable to accept frames.

Bit 3: Ready to Communicate (RTC). The bit is set to 1 when the device is ready to communicate.

Bit 4: Ready to Receive (RTR). The bit is set to 1 when the device is ready to receive data.

Bit 5: Reserved for future use. Set to zero by the sender and ignored by the receiver.

Bit 6: Reserved for future use. Set to zero by the sender and ignored by the receiver.

Bit 7: Incoming Call Indicator (IC). The bit is set to 1 to indicate an incoming call.

Bit 8: Data Valid (DV). The bit is set to 1 to indicate that valid data are being sent.

The control byte is mapped to V.24 signals according to the two tables below:

Table 18: Control Signal Octet Mapping – Receiving Entity

Control Signal Byte	DTE Receiving		DCE Receiving	
	Signal	V.24 circuit	Signal	V.24 circuit
3, RTC	DSR	107	DTR	108/2
4, RTR	CTS	106	RFR	133
7, IC	RI	125	Ignored	-
8, DV	DCD	109	Ignored	-

NOTE

Circuit 133, RFR (Ready for Receiving) is commonly assigned to the connector pin that is alternatively used for circuit 105, RTS (Ready to Send).

Table 19: Control Signal Octet Mapping – Sending Entity

Control Signal Byte	DTE Sending		DCE Sending	
	Signal	V.24 circuit	Signal	V.24 circuit
3, RTC	DTR	108/2	DSR	107
4, RTR	RFR	133	CTS	106
7, IC	Always 0	-	RI	125
8, DV	Always 1	-	DCD	109

NOTE

Circuit 133, RFR (Ready for Receiving) is commonly assigned to the connector pin that is alternatively used for circuit 105, RTS (Ready to Send).

If a station is unable to transmit frames because of flow control but wishes to stop accepting further frames itself, it may send frames containing no user data (i.e., only the control signal octet and, optionally, the break signal octet) in order to control signal flow.

4.4. Procedures

4.4.1. DLC Establishment

In most cases the establishment of a DLC will be initiated by the TE; however, it may be initiated by the UE, as well. The action taken by the higher layers of the TE upon DLC establishment initiation by the UE is out of scope of this document.

The station wishing to establish a DLC transmits a SABM frame with the P-bit set to 1. The address field contains the DLCI value of the desired connection. If the responding station is ready to establish the connection, it will reply with a UA frame with the F-bit set to 1. If the responding station is not ready or unwilling to establish the particular DLC, it will reply with a DM frame with the F-bit set to 1.

Once a DLC has been established, both stations are in connected mode for the particular DLC, and the information transfer may commence.

If no UA or DM response has been received after **<T1>**, the initiating station may retransmit the SABM. This action may be repeated until a response is obtained or action is taken by a higher layer.

If no negotiation procedure is used, DLC parameters are the default one.

4.4.2. DLC Release

The release of a DLC may be initiated by either station by the transmission of a DISC frame with the P-bit set to 1. Confirmation of the DLC release is signaled by the other station sending a UA frame with the F-bit set to 1. Once the DLC has been released, the stations enter disconnected mode for that particular DLC.

If the station receiving the DISC command is already in disconnected mode, it will send a DM response.

If no UA or DM response has been received after **<T1>**, the initiating station may retransmit the DISC. This action may be repeated until a response is obtained or action is taken by a higher layer.

4.4.3. Information Transfer

4.4.3.1. Information Data

In Quectel multiplexing system, data are conveyed using UIH frames. UIH frames may also be used for data transmission when the delays inherent in error-recovery procedures are unacceptable, such as voice data transmission.

The transmitter takes information from the convergence layer for the particular DLC and places it in the information field of the transmitted frame. Once a UIH frame has been correctly received, the contents of its information field are passed to the convergence layer.

The C/R bit of frames sent by the initiating station are set to 1 and C/R bit of those sent by the responding station are set to 0. The P-bit of both stations is set to 0. See **Chapter 4.1.2.2** for more information about the C/R bit.

The maximum length of the information field in UIH frames is **<N1>** in **AT+CMUX**.

4.4.4. Time-out Considerations

In order to detect a no-reply or lost-reply condition, each station provides a response time-out function (<T1>). The expiry of the time-out function is used to initiate appropriate error recovery procedures.

To avoid contentious situations, the duration of the time-out function in the two stations is unequal.

The time-out function is started whenever a station has transmitted a frame for which a reply is required. When the expected reply is received, the time-out function is stopped. The time-out function has to be restarted if other frames requiring acknowledgements are sent during the interval that the time-out function is running.

If the response time-out function runs out, a command with the P-bit set to 1 may be (re)transmitted, and the response time-out function will be restarted.

4.4.5. Flow Control

Quectel multiplexing system supports software flow control mechanism. Software flow control is implemented by 3GPP TS 27.010 MSC, FCoff and FCon message frames.

TE sends MSC message to UE with FC bit set to 1 in V.24 control signals when TE refuses to accept frames, and with FC bit set to 0 to notify recovery of received frames. When receiving MSC, UE sends a response to indicate recovery of data transmission.

TE sends FCoff command to UE when it refuses to accept anything except the control messages on DLCI 0. In such a case, UE stops sending any frame through all data channels except the control channel, which is still alive and free to send any control message. TE can send FCon command to UE to recover the transmission. When UE receives FCoff or FCon command, it sends a response back.

The difference between MSC and FCon is that the former only controls one of the data channels, while the latter controls all the data channels except the control channel.

5 Examples

5.1. Frame Structure Samples

Sample 1:

Opening Flag	Address Field	Control Field	Length Field	FCS	Closing Flag
F9	03	3F	01	1C	F9
Header	DLCI 0	SABM Frame	0, No Information Field		Tail

This sample is a SABM frame to open DLCI 0.

Sample 2:

Opening Flag	Address Field	Control Field	Length Field	Information Field	FCS	Closing Flag
F9	05	EF	09	41 54 49 0D	58	F9
Header	DLCI 1	UIH Frame	4	AT command string "ATI<CR>"		Tail

This sample is a UIH frame to transmit the AT command string "ATI<CR>".

Sample 3:

Opening Flag	Address Field	Control Field	Length Field	Information Field	FCS	Closing Flag
F9	01	EF	0B	E3 07 07 0D 01	79	F9
Header	DLCI 0	UIH Frame	5	MSC Message. Length: 3 octets		Tail

This sample is an MSC message carried in UIH frame to transmit V2.4 signal 0x0D.

5.2. Channel Establishment

Step 1: Launch Multiplexer

No.	Step	Data Direction	Hex	Comment
1	TE launches UE multiplexer function via AT command	TE → UE	61 74 2B 63 6D 75 78 3D 30 0D 0D 0A 4F 4B 0D 0A 0D 0A	AT+CMUX=0<CR><LF>
	UE sends a response	TE ← UE	61 74 2B 63 6D 75 78 3D 30 0D 0D 0A 4F 4B 0D 0A 0D 0A	AT+CMUX=0<CR><LF>OK<CR><LF><CR><LF>

Step 2: Establish DLCI 0

No.	Step	Data Direction	Hex	Comment
1	TE requests to establish control channel DLCI 0 by sending SABM frame	TE → UE	F9 03 3F 01 1C F9	SABM frame
	UE replies with the UA response upon receiving SABM frame and accepts to create DLCI 0	TE ← UE	F9 03 73 01 D7 F9	UA frame

Step 3: Establish DLCI 1–4

No.	Step	Data Direction	Hex	Comment
1	TE requests to establish DLCI 1 by sending SABM frame	TE → UE	F9 27 3F 01 DE F9	
	UE replies with the DM response upon receiving SABM but refuses to create DLCI 1	TE ← UE	F9 27 1F 01 F4 F9	
2	TE requests to establish DLCI 1 by sending SABM frame	TE → UE	F9 07 3F 01 DE F9	
	UE replies with the UA response upon receiving SABM and accepts to create DLCI 1	TE ← UE	F9 07 73 01 15 F9	
	UE sends MSC message frames	TE ← UE	F9 01 EF 0B E3 07 07 0D 01 79 F9	
	UE sends OK	TE ← UE	F9 05 EF 0D 0D 0A 4F 4B 0D 0A 5F F9	
	TE sends MSC message frames	TE → UE	F9 01 EF 0B E3 07 07 0D 01 79 F9	

	UE replies with a response	TE ← UE	F9 01 EF 0B E1 07 07 0D 01 79 F9
	TE requests to establish DLCI 2 by sending SABM frame	TE → UE	F9 0B 3F 01 59 F9
	UE replies with the UA response upon receiving SABM and accepts to create DLCI 2	TE ← UE	F9 0B 73 01 92 F9
3	UE sends MSC message frames	TE ← UE	F9 01 EF 0B E3 07 0B 0D 01 79 F9
	TE sends MSC message frames	TE → UE	F9 01 EF 0B E3 07 0B 0D 01 79 F9
	UE replies with a response	TE ← UE	F9 01 EF 0B E1 07 0B 0D 01 79 F9
4	Establishment of DLCI 3 and DLCI 4 are the same as above.		
5	By now, 4 channels have come into existence, and Quectel multiplexing system can work normally.		

This sample is an MSC message carried in UIH frame to transmit V2.4 signal 0x0D.

5.3. Frame Transmission

Once the control channel and data channels are established, TE and UE can transmit data to each other through UIH frames.

Frame Transmission:

No.	Step	Data Direction	Hex	Comment
	TE sends the AT command string " ATI<CR> " on DLCI 1	TE → UE	F9 05 EF 09 41 54 49 0D 58 F9	UIH frame
1	UE replies with a response on DLCI 1	TE ← UE	F9 05 EF 09 41 54 49 0D 58 F9 F9 05 EF 9B 0D 0A 53 49 4D 43 4F 4D 5F 4C 74 64 0D 0A 53 49 4D 43 4F 4D 5F 53 49 4D 33 30 30 0D 0A 52 65 76 69 73 69 6F 6E 3A 53 49 4D 33 30 30 4D 33 32 28 53 50 41 4E 53 49 4F 4E 29 5F 56 31 30 2E 30 2E 38 5F 42 55 49 4C 44 30 33 0D 0A 0D 0A 47 F9	UIH frame

			F9 05 EF 09 4F 4B 0D 0A 58 F9	
2	TE sends the AT command string "AT<CR>" on DLCI 2	TE → UE	F9 09 EF 07 41 54 0D 35 F9	UIH frame
	UE replies with a response on DLCI 2	TE ← UE	F9 09 EF 07 41 54 0D 35 F9 F9 09 EF 0D 0D 0A 4F 4B 0D 0A D8 F9	UIH frame
3	Frame transmission on DLCI 3 and DLCI 4 are the same as above.			

5.4. Flow Control

Flow Control:

No.	Steps	Data Direction	Hex	Comment
1	UE sends MSC message with FC bit set to 1 on control channel DLCI 0, to indicate refusing to accept anything on DLCI 1.	TE ← UE	F9 01 EF 0B E3 07 07 8F 01 79 F9	
2	UE sends MSC message with FC bit set to 0 on control channel DLCI 0, to indicate recovery of DLC1 data transmission.	TE ← UE	F9 01 EF 0B E3 07 07 8D 01 79 F9	
3	TE sends MSC message with FC bit set to 1 on control channel DLCI 0, to indicate refusing to accept anything on DLCI 1.	TE → UE	F9 01 EF 0B E3 07 07 8F 01 79 F9	
4	TE sends MSC message with FC bit set to 0 on control channel DLCI 0, to indicate recovery of DLC1 data transmission.	TE → UE	F9 01 EF 0B E3 07 07 8D 01 79 F9	
5	TE sends FCoFF message on DLCI 0, to indicate refusing to accept anything on all DLCs except DLCI 0.	TE → UE	F9 01 EF 05 63 01 93 F9	
6	TE sends FCon message through DLCI 0, to indicate recovery of data transmission.	TE → UE	F9 01 EF 05 A3 01 93 F9	

5.5. Close-down of Multiplexer

Close-down Procedures:

No.	Step	Data Direction	Hex	Comment
1	TE sends DISC frame to request for closing down DLCI 1	TE → UE	F9 07 53 01 3F F9	
	UE replies with the UA frame to accept the request	TE ← UE	F9 07 73 01 15 F9	
2	TE sends DISC frame to request for closing down DLCI 2	TE → UE	F9 0B 53 01 B8 F9	
	UE replies with the UA frame to accept the request	TE ← UE	F9 0B 73 01 92 F9	
3	TE sends DISC frame to request for closing down DLCI 3	TE → UE	F9 0F 53 01 7A F9	
	UE replies with the UA frame to accept the request	TE ← UE	F9 0F 73 01 50 F9	
4	TE sends DISC frame to request for closing down DLCI 4	TE → UE	F9 13 53 01 77 F9	
	UE replies with the UA frame to accept the request	TE ← UE	F9 13 73 01 5D F9	
5	TE sends CLD message frame to request for closing down multiplexer on DLCI 0	TE → UE	F9 03 EF 05 C3 01 F2 F9	
	UE acknowledges the CLD message to accept the request	TE ← UE	F9 03 EF 05 C1 01 F2 F9	
6	By now, the multiplexer is closed down.			

6 Appendix References

Table 20: Related Document

Document Name
[1] Quectel_ECx00M&ECx00N&EC800K_Series_AT_Commands_Manual

Table 21: Terms and Abbreviations

Abbreviation	Description
ABM	Asynchronous Balanced Mode
CLD	Multiplexer Close Down
C/R	Command/Response
DCE	Data Communications Equipment
DISC	Disconnect
DLC	Data Link Connection
DLCI	Data Link Connection Identifier
DM	Disconnected Mode
DTE	Data Terminal Equipment (typically computer, MCU, external controller)
DV	Data Valid
FC	Flow Control
FCS	Frame Check Sequence
FCoff	Flow Control off Command
FCon	Flow Control on Command
IC	Incoming Call Indicator

MSC	Modem Status Command
MUX	Multiplexer
PSC	Power Saving Control
RFR	Ready for Receiving
RTC	Ready to Communicate
RTR	Ready to Receive
RTS	Ready to Send
SABM	Set Asynchronous Balanced Mode
TE	Terminal Equipment
UA	Unnumbered Acknowledgement
UE	User Equipment
UI	Unnumbered Information
UIH	Unnumbered Information with Header Check